Date: 06 March 2026

Project Name: UCab - Modern Ride-Sharing Platform

Maximum Marks: 2 Marks

Project Design Phase

Proposed Solution Template

Proposed Solution Template: Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Passengers face long wait times, unpredictable surge pricing, and lack of transparent tracking. Meanwhile, drivers suffer from high platform commission rates (25-30%) and inefficient ride allocations, leading to lower daily earnings.
2.	Idea / Solution description	**UCab** is a full-stack ride-sharing web application built on the **MERN stack** (MongoDB, Express.js, React, Node.js). It features secure JWT authentication, role-based dashboards (Rider/Driver), and real-time communication using **Socket.io**. For location visualization and tracking, it integrates **Leaflet.js**, a free and open-source mapping library, ensuring a lightweight and cost-effective map interface. Payments are handled via a direct UPI QR-code interface.
3.	Novelty / Uniqueness	Unlike traditional apps that rely on expensive proprietary APIs (like Google Maps) and heavy payment gateway fees, UCab utilizes **Leaflet.js** for open-source mapping and a **Peer-to-Peer (P2P) UPI QR Code** system for payments. This drastically reduces the operational cost of the platform and eliminates middleman gateway commissions.

4.	Social Impact / Customer Satisfaction	By reducing platform overhead costs, UCab provides **cheaper rides for passengers** and **higher take-home earnings for drivers**. The intuitive, mobile-responsive UI (built with Vite & Bootstrap) ensures a seamless and satisfying booking experience for users of all age groups.
5.	Business Model (Revenue Model)	Instead of the traditional high-percentage commission model, UCab operates on a **Zero-Commission / SaaS Model**. The platform generates revenue by charging drivers a small, flat daily/monthly subscription fee to access the network, or by charging a nominal flat "convenience fee" (e.g., ₹5) per ride from the passenger.
6.	Scalability of the Solution	The architecture is highly scalable. The **Node.js/Express** backend uses stateless **JWT authentication**, allowing horizontal scaling across multiple servers. **MongoDB** efficiently handles large volumes of ride data. **Socket.io** manages concurrent real-time tracking connections, while **Leaflet.js** offloads map rendering to the client's browser, keeping server loads minimal.